

Lab Class NLP:III**Exercise 1 : Zipf Distribution**

The lecture introduced Zipf's Law as a statistical law that describes the frequency of words in a language.

- (a) What does Zipf's law state?
- (b) What implications does it have for developing statistical models in NLP?
- (c) Consider the following table of words that appeared in a corpus:

Rank	Word	Frequency	Expected Frequency
1	the	36000	
2	that	18000	
3	for	12000	
4	is	9000	
5	said	7200	
6	on	6000	
8	in	4500	
9	it	4000	
10	by	3600	
12	from	3000	
15	million	2400	
16	at	2250	
18	as	2000	
20	with	1800	
24	a	1500	
25	was	1440	

Assuming that there are a total of 500,000 words out of which 50,000 are unique, do the word frequencies satisfy Zipf's law in this case? Explain why or why not.

Exercise 2 : Data Acquisition

You decide to study the problem of computational argumentation.

- (a) Name two possible sources of argumentative text like arguments, debates, or claims as text or speech transcripts.
- (b) For each source from (a), describe how you would collect this data and what problems you could face collecting them.
- (c) Inform yourself about common sources of bias in language corpora or datasets. Name the biases you found and how they can be mitigated during data acquisition.

Exercise 3 : Corpora and Annotation

In this exercise, you will explore the challenges involved in annotating, both for humans and automated systems. Consider the following corpus:

1. Paris Hilton stayed at the Hilton in Paris.
2. Quentin Bloody Tarantino.
3. Max works for Berlin & Brandenburg Post.

(a) Identify and annotate all named entities in the sentences above. For this task, a named entity is any word or phrase that refers to

- a person (PER),
- a location (LOC), or
- an organisation (ORG).

Use the following format for your annotations: [Mary Smith]_{PER} went to [Tokyo]_{LOC}.

(b) Discuss any instances you found challenging to annotate. Which cases might be particularly difficult for an automated system to annotate? Why?